

It is evident that G_1 and G_2 are Lipschitz continuous on any bounded set. From Lemma 2(3), we know that $\mathcal{W}^t$ is bounded. Hence, we obtain

$$\|V^{t+1}\| \leq \varpi \|\mathcal{W}^{t+1} - \mathcal{W}^t\|, \quad (3.33)$$

where $\varpi = \max_{i \in [6]} \{\rho_i + l_g\}$ and l_g is the maximum Lipschitz constant of G_1 and G_2 . $\square$

Lemma 4 *The function $F(\mathcal{W})$ is a Kurdyka-Łojasiewicz (KL) function.*

Proof According to [5], the Frobenius norm functions $\frac{\lambda_1}{2}\|B\|^2$, $\frac{\lambda_3}{2}\|\mathcal{H} - \mathcal{C} \times_3 B - \mathcal{E}_1\|^2$, and $\frac{\lambda_6}{2}\|\mathcal{C} - \mathcal{D} * \mathcal{Z}^T - \mathcal{E}_2\|^2$ are semi-algebraic functions. The terms $\lambda_2\|\mathcal{E}_1\|_{11\phi}$ and $\lambda_5\|\mathcal{E}_2\|_{11\phi}$ are also semi-algebraic functions because they are composed of semi-algebraic operations, including the Frobenius norm, minimum, and finite summation. Similarly, $\lambda_4\|\mathcal{Z}\|_{F,1}^\Psi$, which involves only the Frobenius norm and finite summation, is semi-algebraic. The functions $\delta_{\mathbb{B}}(B)$, $\delta_{\mathbb{C}}(\mathcal{C})$, and $\delta_{\mathbb{D}}(\mathcal{D})$ are semi-algebraic because they are indicator functions with semi-algebraic sets [5]. Hence, $F(\mathcal{W})$ is semi-algebraic because it is a finite sum of semi-algebraic functions. Additionally, since $F(\mathcal{W})$ is a proper continuous function, it follows from [5, Theorem 3] that $F(\mathcal{W})$ is a KL function. This completes the proof. $\square$

Finally, we provide a theoretical guarantee for the convergence of Algorithm 1.

Theorem 4 *Consider the sequence $\{\mathcal{W}^t\}$ obtained by Algorithm 1. Assuming that either $\mathcal{Z}^t$ or $\mathcal{E}_2^t$ is bounded, the sequence $\{\mathcal{W}^t\}$ converges to a critical point of $F(\mathcal{W})$.*

Proof We begin by noting that Lemma 2(3) establishes the boundedness of the sequence $\{\mathcal{W}^t\}$. Given this boundedness, the Bolzano-Weierstrass theorem ensures the existence of a convergent subsequence. Furthermore, by exploiting the continuity of $F(\mathcal{W})$, along with the results derived from Lemmas 2, 3, 4, we can rigorously establish the desired conclusion as articulated in [2, Theorem 2.9]. $\square$

3.7. Rank reduction strategy with validation mechanism

In this subsection, we propose a rank reduction strategy to decrease r , which is related to the dimensions of $\mathcal{D} \in \mathbb{R}^{n_1 \times r \times b}$ and $\mathcal{Z} \in \mathbb{R}^{n_2 \times r \times b}$, thereby reducing the complexity of Algorithm 1. Unlike most decomposition methods that set a small initial r^0 , we set $r^0 = \min\{n_1, n_2\}$ and leverage the algorithm's inherent capability to adaptively reduce r^t . This approach is initially supported by Theorem 2. Furthermore, we present the following theorem to substantiate our proposed rank reduction strategy.

Theorem 5 *There exists $t^\# \in \mathbb{N}$ such that $\Gamma(\mathcal{Z}^t) = \Gamma(\mathcal{Z}^{t+1})$ for each $t \geq t^\#$, where $\Gamma(\mathcal{Z}) = \{j \mid \|\mathcal{Z}(:, j, :)\| \neq 0, j = 1, \dots, r\}$.*

Proof By Lemma 2 that $\lim_{t \rightarrow \infty} \|\mathcal{Z}^{t+1} - \mathcal{Z}^t\| = 0$, there exists $t^\# \in \mathbb{N}$ such that $\|\mathcal{Z}^{t+1} - \mathcal{Z}^t\| < \min\{(2\hat{\lambda}_4(1-p))^{1/(2-p)}, \nu\}$ for each $t \geq t^\#$. Proving by contradiction, we assume that there exists $t \geq t^\#$ such that $\Gamma(\mathcal{Z}^t) \neq \Gamma(\mathcal{Z}^{t+1})$. Then there exists $j \in r$ such that (a) $\mathcal{Z}^t(:, j, :) \neq 0$ and $\mathcal{Z}^{t+1}(:, j, :) = 0$, or (b) $\mathcal{Z}^t(:, j, :) = 0$ and $\mathcal{Z}^{t+1}(:, j, :) \neq 0$. Hence by (3.19), we have for both cases that

$$\|\mathcal{Z}^{t+1} - \mathcal{Z}^t\| \geq \|\mathcal{Z}^{t+1}(:, j, :) - \mathcal{Z}^t(:, j, :)\| \geq \min\{(2\hat{\lambda}_4(1-p))^{1/(2-p)}, \nu\},$$

which yields a contradiction. The proof is complete. $\square$